

Cardiac Responses to Daily Threats and Challenges During Wakefulness and Sleep

William von Hippel¹, Jeongeun Kim², Finnbar Fielding², Christopher J. Chapman², Emily R. Capodilupo³, Gregory J. Grosicki², and Kristen E. Holmes^{2, 4}

¹ Research with Impact, Chapel Hill, Queensland, Australia

² Department of Performance Science, WHOOP Inc., Boston, Massachusetts, United States

³ Department of Data Science and Research, WHOOP Inc., Boston, Massachusetts, United States

⁴ School of Psychology, University of Queensland

This research examines cardiovascular response to everyday threats and challenges during wakefulness and sleep. Approximately 11,000 people, who comprised a diverse sample ethnically but not socioeconomically, completed three weekly morning and evening surveys in which they indicated whether they expected and experienced threats and challenges that day. Participants also provided measures of blood pressure on morning surveys and provided measures of average heart rate during the day and resting heart rate when asleep via their WHOOP biometric capture device. Enrollment began in April 2024, and data collection ceased in July 2024. Results indicated that both threat and challenge were associated with higher blood pressure and higher average heart rate during the day. In contrast, when people were asleep, threat was associated with higher resting heart rate but challenge was associated with lower resting heart rate. These results suggest that the body achieves more restorative sleep in preparation for perceived challenges but not for perceived threats, raising the possibility that the greater stress associated with threats disrupts the body's capacity for restorative sleep. The generalizability of these results to members of economically marginalized groups remains an open question.

Keywords: threat, challenge, heart rate, blood pressure, sleep

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Daily life is replete with stressors. Sometimes stressors threaten to overwhelm people, due to lack of resources, time, or the opportunity to address them. In such cases, stressors are appraised primarily as a threat (Lazarus & Folkman, 1984). In contrast, sometimes people are prepared for stressors, having the resources, time, and opportunity to address them successfully. In such cases, stressors are appraised primarily as a challenge (Lazarus & Folkman, 1984). Stressors that are appraised as threatening typically lead to larger stress responses (e.g., greater increases in cortisol; Denson et al., 2009), and there are stable individual differences in the tendency

to appraise events as challenging versus threatening (Tomaka & Magoc, 2021). Challenge versus threat appraisals influence people's coping strategies (Hulin et al., 2024) and play a mediating role in whether stressors lead to negative outcomes such as intentions to leave one's employer or career (Webster et al., 2011).

Threats and challenges are both experienced as subjectively stressful, given that a positive outcome is not guaranteed in either case, but people tend to seek out and enjoy challenges but avoid and dislike threats (Tomaka et al., 1993). Perhaps the most commonplace way that people seek challenges and avoid threats is through

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Keely Muscatell served as action editor.

William von Hippel  <https://orcid.org/0000-0001-9900-0868>

All code used in the analyses is available in the [Supplemental Materials](#). The data can be provided by Kristen E. Holmes pending scientific review. Requests for the data should be submitted to Kristen E. Holmes. All authors work directly or indirectly for WHOOP and hence could benefit financially by research findings that are of value to WHOOP. Because neither WHOOP nor the authors have any vested interest in the particular outcome of this research, however, there are no conflicts of interest regarding the pattern of findings. There is no external funding—all costs are paid by WHOOP.

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Correspondence concerning this article should be addressed to Kristen E. Holmes, Department of Performance Science, WHOOP Inc., 1 Kenmore Square, Boston, MA 02151, United States. Email: holmes@whoop.com

learning and deliberate practice. By increasing their capabilities and resources, people turn anticipated threats into challenges and thereby manage their environment (Brinums et al., 2023).

Learning and practice can offset the risks and maximize the benefits associated with threats and challenges over the long term, but how does the body prepare for threats and challenges in the short term? According to the Biopsychosocial Model of Threat and Challenge (Blascovich & Tomaka, 1996; Seery, 2013), one way the body mobilizes energy to engage with threats and challenges is by increasing heart rate. Increased heart rate may or may not result in increased blood pressure, depending on whether it is accompanied by a reduction in total peripheral resistance brought about by vasodilation. The Biopsychosocial Model proposes that challenge leads to greater vasodilation than threat, thereby resulting in a smaller increase in blood pressure.

Research provides general support for these tenets of the Biopsychosocial Model. For example, when people are faced with stressors, the ratio of demands to resources is a stronger predictor of increases in blood pressure than demands alone (Gordon & Mendes, 2021). These results indicate that people experience a greater increase in blood pressure when threatened (when demands exceed resources) than when challenged (when resources exceed demands). Consistent with this interpretation, threat and challenge appraisals can be differentiated by their distinct cardiac profiles (Khalaf et al., 2020).

Although these findings and many others like them speak to how the body prepares for threat and challenge when awake, they do not address the question of what happens during sleep. On the one hand, the body may respond during sleep as it does when awake, with an increase in resting heart rate in preparation for anticipated threats and challenges. On the other hand, people cannot actively deal with stressors when sleeping, so an increase in heart rate during sleep would achieve little. Additionally, sleep is more restorative when people are in a state of relaxation and parasympathetic dominance (Trinder et al., 2001), with a commensurate decrease in resting heart rate. For these reasons, when anticipating threats and challenges, people may show an increase in heart rate when awake but no change or even a decrease in heart rate when sleeping.

Because stressful events can disrupt sleep (Kalmbach et al., 2018; Kim & Dimsdale, 2007), however, anticipated threats might be antithetical to the relaxation required for restful sleep. As a result, threats (and possibly challenges as well) might increase heart rate during sleep and thereby disrupt the body's capacity to achieve restorative sleep. Consistent with this possibility, sleep disturbances tend to be bidirectionally related to stress and anxiety, which are exacerbated when people appraise their circumstances as threatening (Jenkins et al., 2024; Spoormaker & van den Bout, 2005).

The goal of the current research was to compare cardiac responses to threats and challenges in the context of people's everyday lives when they are awake versus asleep. In service of this goal, we conducted an exploratory study in which we measured anticipated threat and challenge in three weekly morning surveys and retrospectively assessed threat and challenge in three weekly evening surveys (administered on different days). We invited members of the WHOOP platform to participate in our research, which enabled us to collect measures of heart rate during sleep and wakefulness via biometric capture.

To differentiate the effects of threat and challenge appraisals from acute feelings of stress, we also asked people in the morning and

evening surveys how stressed and anxious they were feeling at the moment. Finally, during the morning survey, we asked people to collect and report their blood pressure using a blood pressure cuff. Approximately 11,000 people completed at least three morning and three evening surveys and provided three measures of blood pressure, enabling modeling of the within- and between-person relationships between threat, challenge, acute stress, heart rate, and blood pressure.

We used these data to address four research questions:

1. Do people report higher levels of acute stress when they face a challenge than when they do not and even higher levels of acute stress when they face a threat?
2. Do people experience higher average heart rates during wakefulness when they face challenges or threats compared to when they face neither?
3. Do people experience higher blood pressure when they face a challenge than when they do not and even higher blood pressure when they face a threat?
4. Do people show an increase, decrease, or no change in resting heart rate during sleep when they anticipate a challenge or threat the next day compared to when they anticipate neither?

Method

Participants

Sample size was determined by the number of WHOOP members who volunteered to participate, with the minimum *N* set at 10,000. The study was advertised through WHOOP marketing channels to clients in the United States. Participants joined the study in return for free replacement bands for their WHOOP device, personalized reports of their results, and the opportunity to be entered into a raffle for prizes such as a one-to-one coaching session with WHOOP staff. Enrollment began in April 2024, and data collection ceased in July 2024. A total of 7,156 males and 3,882 females residing in the United States completed at least three morning and three evening surveys and provided blood pressure measures. The sample had an age range from 18 to 84, a mean age of 40.35, and an average body mass index (BMI) of 26.61 (*SD* = 4.76). A total of 8,780 members of the sample identified as White/Caucasian, 805 as Hispanic, 764 as Asian, 376 as Black/African American, 86 as American Indian/Alaska Native, 40 as Native Hawaiian, and 135 as "Other," and the remainder indicated that they preferred not to say. Ethics permission was provided by Salus Institutional Review Board, Number 24238.

Design

The objectives of the current research were to examine the relationships between threat and challenge, stress, blood pressure, and heart rate during wakefulness and sleep. In service of these goals, members of the WHOOP platform were invited to complete three morning surveys and three evening surveys. Participants were invited to complete each morning and evening survey at 1-week intervals, with the evening survey invitation sent the day prior to the morning survey invitation.

Morning Survey

In the context of a larger set of questions that asked about a variety of unrelated issues, participants were asked about anticipatory threat and challenge with the following question: “Thinking about the day ahead of you, do you expect to face any of the following?” They were then provided with tick boxes that were labeled “Challenges (opportunities to test yourself for which you are prepared and can succeed),” “Threats (events that could overwhelm you and/or where success is unlikely due to lack of opportunity, resources, or skills),” or “Neither.” Participants could tick boxes for both threats and challenges. Participants were also asked to indicate their current stress level with the question “How would you rate your level of stress or anxiety right now?” with responses provided on a 5-point scale ranging from 1 = *extremely stressed* to 5 = *not at all stressed* (reverse scored).

At the time of the morning survey, participants were also asked to measure their blood pressure three times using a blood pressure cuff, following American Heart Association guidelines in which each reading is taken 1 min apart (Pickering et al., 2005). Blood pressure was recorded anywhere from 30 min to 2 hr after people woke up in the morning, while they were seated, and prior to eating or taking any medication. Participants were asked to provide systolic and diastolic measures for all readings. We report the average of these three readings as the measure of blood pressure in our analyses, but the results were nearly identical when we used any of the three individual readings as our measure of blood pressure (note that systolic blood pressure increased by 1.2 points across the three readings and diastolic blood pressure increased by .2 points; see [Supplemental Materials](#) for the results of these analyses).

Evening Survey

In the context of a larger set of questions that asked about a variety of unrelated issues, participants were asked about retrospective threat and challenge with the following question: “Thinking about your day today, did you face any of the following?” They were then provided with tick boxes with the same *challenge*, *threat*, and *neither* labels from the morning survey. As with the morning survey, participants could tick boxes for both threats and challenges. Participants were also asked to indicate their current stress level with the same question from the morning survey, with responses provided on a 5-point scale ranging from 1 = *not at all stressed* to 5 = *extremely stressed*.

Heart Rate While Sleeping and Awake

Heart rate was measured passively by the WHOOP biometric capture device (WHOOP 3.0, Inc., Boston, Massachusetts), which underestimates heart rate by .3 beats per minute and has an intraclass correlation of .99 when compared with electrocardiogram-derived measurement of heart rate (Miller et al., 2022). Heart rate during the day was computed as the average heart rate over the course of people’s waking hours, including times when they are moving (the biometric capture device can detect movement across all three axes). The WHOOP device is capable of measuring heart rate accurately even when people are in motion, through the use of proprietary algorithms that remove motion artifacts. In situations where motion artifacts cannot be removed, WHOOP has a mechanism for flagging

low confidence in the heart rate estimate due to high levels of movement, and the algorithm then smooths or interpolates over brief periods of inaccuracy.

Heart rate during sleep was calculated as a weighted average that emphasizes periods later at night and episodes of slow wave sleep, when the body is most at rest and bodily movements are minimized (Brandenberger et al., 2005). This strategy allows for the clearest comparison of one night to the next, independent of how often participants wake up, how active they are while dreaming, and so forth. Our use of a biometric capture device to measure heart rate during the day and while people were sleeping had the advantage of allowing us to collect data from a large sample of people in the context of their everyday lives, but the disadvantage that we did not have the control and measurement precision afforded in laboratory studies (see Quigley et al., 2024).

Statistical Analysis

Due to the nested structure of the data, whereby all measures were repeated within individuals, analyses were conducted using multilevel models (in SPSS 29.02). Within-person effects were separated from between-person effects via person mean centering by computing each individual’s mean to assess between-person effects and computing the daily difference from that individual mean to assess within-person effects. Variance inflation factors ranged between 1.01 and 1.17 across the different models, suggesting there were no concerns with multicollinearity. Because this study was exploratory and not preregistered, all analyses were conducted in multiple ways to ensure consistency and robustness of findings.

Specifically, we used both fixed and random slopes and we also included each predictor variable on its own in the model as well as in combination with the other predictors in a single model. Additionally, we conducted the analyses with and without the control variables of sex, BMI, and age. The results were largely equivalent regardless of analytic approach, but we note when effects did not emerge consistently across models (and we report the results of all models in the [Supplemental Materials](#)). All analyses are reported here with random intercepts, random slopes, and controls for sex, age, and BMI.

In the final set of analyses we report in the article, we used linear probability models to estimate the degree to which heart rate while sleeping predicted threat and challenge appraisals the next day. Although threat and challenge appraisals are binary outcomes, their probability in our sample was linearly related to the log odds function used in logistic regression. In such circumstances, linear models have the advantage of ease of interpretation while fitting the data just as well as the logistic model (e.g., Gomila, 2021).

Transparency and Openness

The full results of all analyses and the analytic syntax are provided in the [Supplemental Materials](#), where they follow the same structure as the article itself. Additionally, all relevant measures are reported herein and no data were removed prior to analyses. In sum, we report how we determined our sample size, all data exclusions (if any), all manipulations, and all measures in the study. Finally, we have not submitted any additional papers that analyze the threat/challenge data, and all subsequent publications that might be completed with these threat/challenge data will reference this article. The

data will be made available upon request to the corresponding author pending scientific review.

Results

Table 1 presents the means and standard deviations of the various measures. Table 2 presents the total number of anticipated and retrospective threats and challenges.

Question 1: The Relationship Between Threat, Challenge, and Acute Stress

The first step in our analyses was to examine the relationship between threat, challenge, and acute stress. According to predictions, threat should be a more stressful state than challenge. As can be seen in Figure 1, that prediction was born out; stress ratings were higher on days when participants reported challenge compared to no challenge and higher still on days when participants reported threat.

To separate the within- from between-person components of these relationships, we estimated a multilevel model with random slopes and intercepts, acute stress as the dependent variable, threat and challenge as predictor variables, and controls for age, sex, and BMI. The model revealed that both threat and challenge were associated with greater within- and between-person feelings of acute stress. The within-person coefficients indicate that people felt more stressed on days when they felt threat or challenge compared to days when they did not. The between-person coefficients indicate that people who tend to experience more frequent threat and challenge also experience greater stress than people who tend to experience less frequent threat and challenge. Note that the coefficients were substantially larger for

Table 1
Sample Means and Standard Deviations

Measure	mean	SD
Heart rate during sleep	<i>M</i> = 59.30 <i>SD</i> = 8.64 w <i>SD</i> = 3.83	
Average heart rate while awake	<i>M</i> = 72.90 <i>SD</i> = 6.61 w <i>SD</i> = 3.92	
Systolic blood pressure	<i>M</i> = 115.34 <i>SD</i> = 12.20	
Diastolic blood pressure	<i>M</i> = 73.76 <i>SD</i> = 8.53	
Anticipated challenge (1 = yes, 0 = no)	<i>M</i> = 0.60 <i>SD</i> = 0.37	
Retrospective challenge (1 = yes, 0 = no)	<i>M</i> = 0.54 <i>SD</i> = 0.35	
Anticipated threat (1 = yes, 0 = no)	<i>M</i> = 0.050 <i>SD</i> = 0.15	
Retrospective threat (1 = yes, 0 = no)	<i>M</i> = 0.092 <i>SD</i> = 0.18	
Morning acute stress/anxiety (1–5 scale)	<i>M</i> = 2.16 <i>SD</i> = 0.78	
Evening acute stress/anxiety (1–5 scale)	<i>M</i> = 2.20 <i>SD</i> = 0.81	

Note. w*SD* = within-person standard deviation.

Table 2
Total Number of Threat and Challenge Appraisals

Measure	Yes	No
Anticipated threat	1,888	35,327
Retrospective threat	3,793	37,324
Anticipated challenge	22,362	14,853
Retrospective challenge	21,931	19,186

threat than challenge, with the largest relationship being the between-person effect of threat.

As can be seen in Table 3, challenge is associated with a within-person increase in stress of about one quarter of a point and a between-person increase in stress of approximately half a point. In contrast, threat is associated with a within-person increase in stress of about three quarters of a point and a between-person increase in stress of approximately one-and-three-quarters points. This pattern of relationships between threat, challenge, and stress emerged in nearly identical form across all the various instantiations of the multilevel models (see Supplemental Materials).

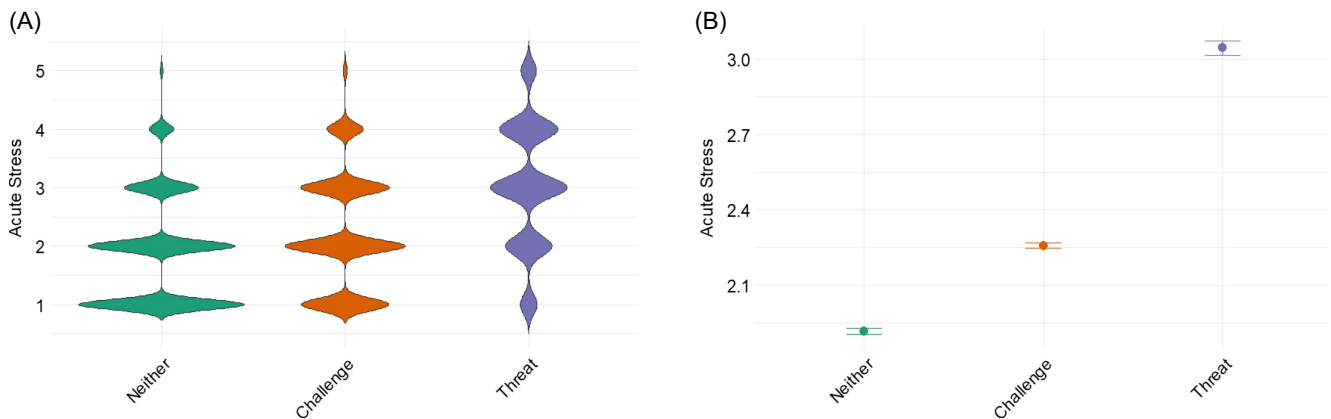
Question 2: Threat, Challenge, and Heart Rate When Awake

Having established that threat was associated with greater stress in our sample than challenge, the next step was to examine the impact of threat, challenge, and stress on daytime average heart rate. According to predictions, average heart rate should increase when people experience threat and challenge, but because heart rate also increases with stress, the effect of threat should be larger than the effect of challenge. As can be seen in Figure 2, these predictions were born out, with increases in heart rate of approximately 1/2 beat per minute (bpm) on days when people experienced challenge and an additional increase of approximately 1/2 bpm on days when they experienced threat. As can be seen in Figure 2, each point on the stress scale from 1 to 5 was also associated with approximately a 1/2 bpm increase as well.

To establish whether threat and challenge are associated with within- and between-person increases in average daily heart rate when controlling for acute stress levels, we estimated a multilevel model with random slopes and intercepts; average daily heart rate as the dependent variable; threat, challenge, and stress as predictor variables; and controls for age, sex, and BMI (see Table 4). The multilevel model revealed that both threat and challenge predicted within-person increases in average heart rate during the day when controlling for stress, although challenge predicted a greater within-person increase in heart rate (.8 bpm) than threat (.5 bpm). This larger within-person effect of challenge than threat emerged reliably across the different instantiations of the multilevel models. These within-person coefficients for threat and challenge indicate that people had faster average heart rates on days when they felt threat or challenge compared to days when they did not. Neither threat nor challenge predicted between-person increases in heart rate in the full model, although threat did in some models (see Supplemental Materials). Acute stress did not account for unique variance in within-person average heart rate beyond threat and challenge, but it reliably predicted between-person increases in average heart rate. The between-person coefficient for acute stress indicates that people

Figure 1

Violin Plots (Panel A) and Mean Estimates With 95% CIs (Panel B) of Acute Stress Levels by Challenge and Threat



Note. Each violin plot represents the density of data points within each group, with increased width reflecting a greater number of observations. Stress levels were self-reported on a scale of 1–5, where higher values indicate greater stress. CI = confidence interval. See the online article for the color version of this figure.

who experienced greater stress had faster average heart rates than people who experienced less stress.

Question 3: Threat, Challenge, and Blood Pressure

The finding that threat and challenge are both associated with increases in daily average heart rate suggests that they are both likely to increase blood pressure as well. Recall, however, that the Biopsychosocial Model predicts that the effects of threat on blood pressure will be larger than that of challenge due to differences in vasodilation. Consistent with that possibility, Figure 3 shows that challenge is associated with about a one quarter point increase in systolic blood pressure and threat is associated with approximately an additional full point increase beyond that (see Figure 3A). Similar, albeit smaller, effects emerged with diastolic blood pressure (see Figure 3B).

To establish whether threat and challenge are associated with within- and between-person increases in blood pressure when controlling for acute stress levels, we estimated multilevel models with random slopes and intercepts; systolic and diastolic blood pressure as the dependent variables; threat, challenge, and stress as predictor variables; and controls for age, sex, and BMI. The multilevel models revealed that challenge showed an unexpected relationship with blood pressure, whereby it predicted a within-person increase of approximately 1/2 point but a between-person decrease of approximately 1 point, in both systolic and diastolic blood pressure (see Table 5). This within-person increase was evident across the various instantiations of the multilevel models, while the between-person decrease emerged in some models but not others (see Supplemental Materials). The within-person effect of challenge indicates that people had higher blood pressure on days when they felt challenge compared to days when they did not.

Table 3

Within- and Between-Person Effects of Challenge and Threat on Acute Stress

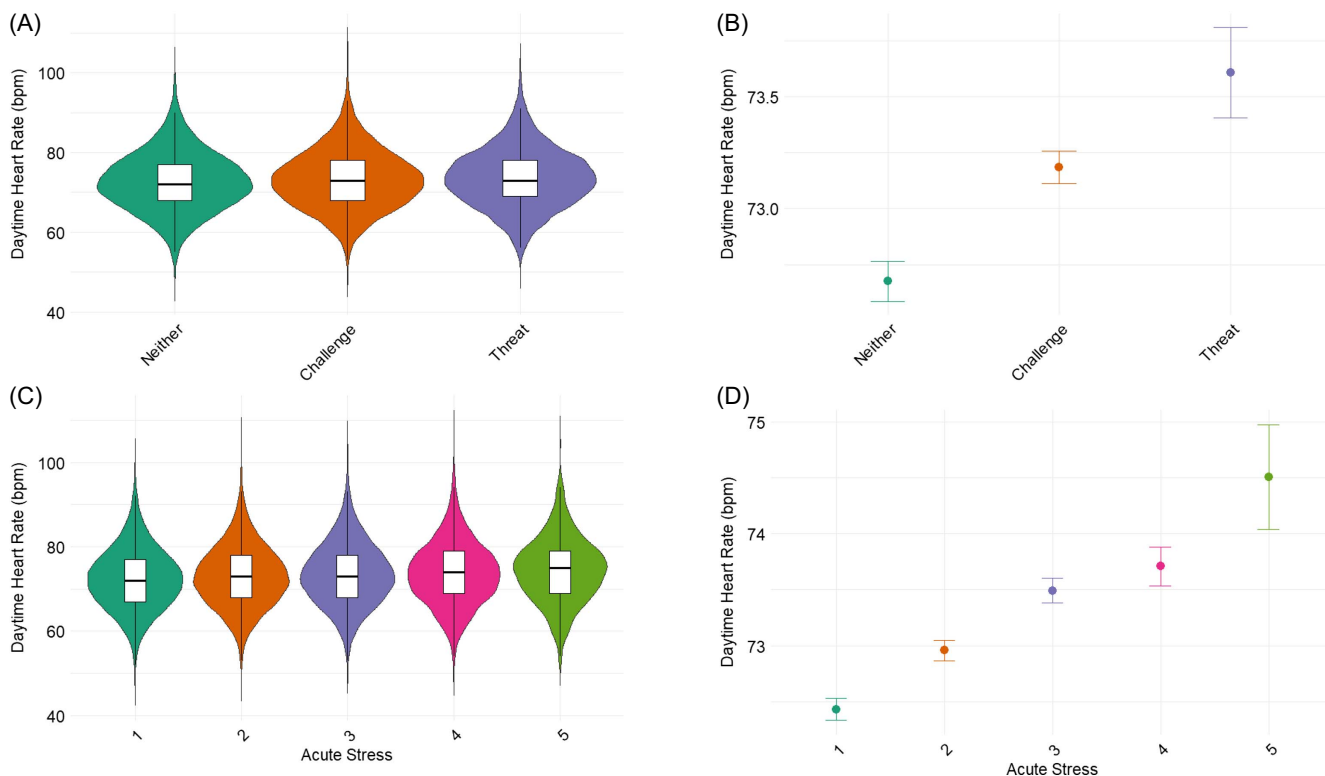
Dependent variable: Acute stress						95% CI	
Parameter	Estimate	SE	df	t	Sig.	LL	UL
Intercept	1.833	0.038	18283.734	48.146	<0.001	1.758	1.908
BMI	0.008	0.001	18385.210	7.045	<0.001	0.006	0.010
Age	−0.007	0.000	18044.599	−14.618	<0.001	−0.008	−0.006
Sex (M = 0, F = 1)	0.095	0.012	17952.161	8.279	<0.001	0.073	0.118
Challenge (W)***	0.224	0.010	55915.296	22.520	<0.001	0.205	0.244
Challenge (B)	0.458	0.017	18769.101	26.762	<0.001	0.425	0.492
Threat (W)***	0.723	0.018	54266.106	40.061	<0.001	0.688	0.759
Threat (B)	1.805	0.037	19536.304	48.651	<0.001	1.732	1.878

Note. Rows with (W) report within-person effects. Rows with (B) report between-person effects. Stress levels were self-reported on a scale of 1–5, where higher values indicate greater stress. Asterisks after within-person effects indicate if the variance of the random slopes is significantly greater than zero, suggesting individual differences in fixed effects. SE = standard error; Sig. = significance; CI = confidence interval; LL = lower limit; UL = upper limit; BMI = body mass index; M = male; F = female.

*** $p < .001$.

Figure 2

Violin Plots of Daytime Average Heart Rate (bpm) by Challenge and Threat (Panel A) and Acute Stress (Panel C)



Note. Each violin plot shows the distribution of heart rate values, with increased width reflecting a greater density of observations. Boxplots within the violins indicate the median and interquartile range of heart rate values for each group. Stress levels were self-reported on a scale of 1–5, where higher values indicate greater stress. Mean estimates with 95% confidence intervals are reported in Panels B and D. See the online article for the color version of this figure.

In contrast, threat showed a within-person increase of one-and-a-quarter points in systolic blood pressure but no other significant relationships. This larger within-person effect of threat than challenge on systolic blood pressure emerged across the various models, but

otherwise, the effects of threat on systolic and diastolic blood pressure varied by model. The within-person effect of threat indicates that people had higher systolic blood pressure on days when they felt threat compared to days when they did not. Finally, acute stress

Table 4

Within- and Between-Person Effects of Challenge, Threat, and Acute Stress on Average Heart Rate

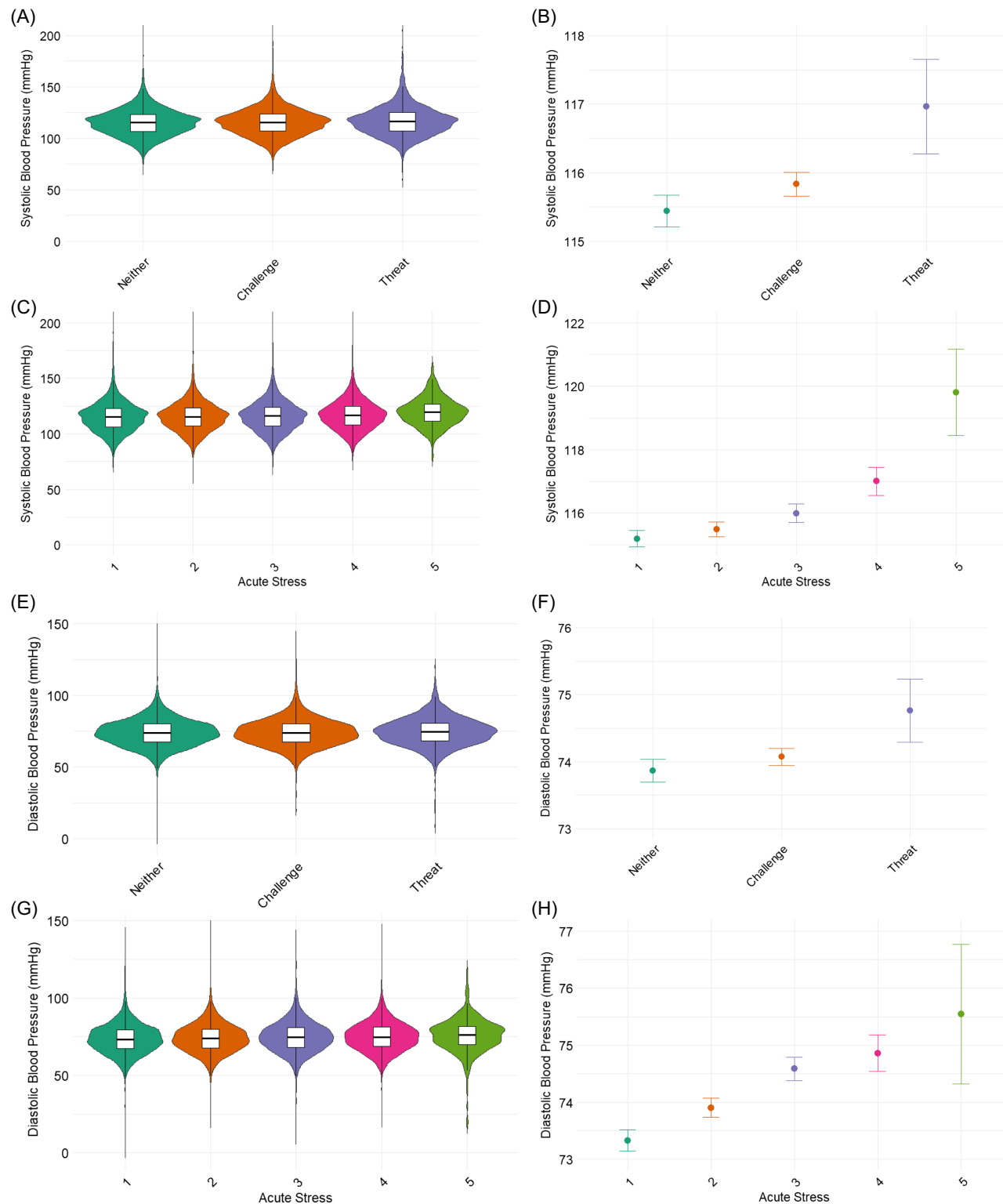
Dependent variable: Daytime average heart rate							
Parameter	Estimate	SE	df	t	Sig.	95% CI	
						LL	UL
Intercept	65.125	0.305	30829.763	213.471	<0.001	64.527	65.723
BMI	0.236	0.008	30868.085	27.776	<0.001	0.219	0.252
Age	−0.020	0.004	30539.251	−5.817	<0.001	−0.027	−0.014
Sex (M = 0, F = 1)	3.195	0.087	30442.100	36.935	<0.001	3.026	3.365
Challenge (W)***	0.811	0.066	56876.788	12.226	<0.001	0.681	0.941
Challenge (B)	−0.094	0.130	31296.617	−0.722	0.470	−0.350	0.161
Threat (W)	0.457	0.124	58114.382	3.676	<0.001	0.213	0.700
Threat (B)	0.325	0.294	32089.307	1.104	0.270	−0.252	0.902
Stress (W)***	0.006	0.049	39365.135	0.115	0.909	−0.091	0.102
Stress (B)	0.647	0.059	31506.959	10.934	<0.001	0.531	0.763

Note. Rows with (W) report within-person effects. Rows with (B) report between-person effects. Asterisks after within-person effects indicate if the variance of the random slopes is significantly greater than zero, suggesting individual differences in fixed effects. SE = standard error; Sig. = significance; CI = confidence interval; LL = lower limit; UL = upper limit; BMI = body mass index; M = male; F = female.

*** $p < .001$.

Figure 3

Violin Plots of Systolic Blood Pressure (Panels A and C) and Diastolic Blood Pressure (Panels E and G) by Challenge and Threat and Acute Stress



Note. Each violin plot shows the distribution of blood pressure values, with increased width reflecting a greater density of observations. Boxplots within the violins indicate the median and interquartile range of blood pressure values for each group. The y-axis is trimmed at 200 mmHg for systolic and 150 mmHg for diastolic blood pressure plots for ease of visualizing effects (for systolic blood pressure, nine outliers were removed; for diastolic blood pressure, seven outliers were removed). Stress levels were self-reported on a scale of 1–5, where higher values indicate greater stress. Mean estimates with 95% confidence intervals are reported in Panels B, D, F, and H. See the online article for the color version of this figure.

Table 5*Within- and Between-Person Effects of Threat, Challenge, and Acute Stress on Blood Pressure*

Parameter	Estimate	SE	df	t	Sig.	95% CI	
						LL	UL
Dependent variable: Systolic blood pressure							
Intercept	94.476	0.588	22404.596	160.537	<0.001	93.322	95.629
BMI	0.747	0.016	22424.077	45.495	<0.001	0.715	0.779
Age	0.077	0.007	22218.293	11.384	<0.001	0.064	0.090
Sex (M = 0, F = 1)	-9.812	0.167	22158.530	-58.723	<0.001	-10.140	-9.484
Challenge (W)	0.476	0.166	35036.668	2.866	0.004	0.150	0.801
Challenge (B)	-0.953	0.250	22932.284	-3.806	<0.001	-1.444	-0.462
Threat (W)***	1.265	0.365	32595.947	3.468	<0.001	0.550	1.980
Threat (B)	0.092	0.583	23257.723	0.158	0.874	-1.051	1.236
Stress (W)***	0.233	0.108	28621.600	2.163	0.031	0.022	0.445
Stress (B)	1.070	0.115	22909.273	9.327	<0.001	0.846	1.295
Dependent variable: Diastolic blood pressure							
Intercept	55.502	0.454	21865.779	122.129	<0.001	54.611	56.393
BMI	0.533	0.013	21884.135	42.072	<0.001	0.508	0.558
Age	0.089	0.005	21669.914	17.097	<0.001	0.079	0.099
Sex (M = 0, F = 1)	-3.459	0.129	21604.845	-26.815	<0.001	-3.712	-3.206
Challenge (W)*	0.425	0.129	35102.897	3.288	0.001	0.172	0.679
Challenge (B)	-0.951	0.193	22423.287	-4.916	<0.001	-1.330	-0.572
Threat (W)***	0.202	0.284	32548.601	0.712	0.477	-0.354	0.758
Threat (B)	-0.503	0.451	22759.382	-1.116	0.264	-1.387	0.380
Stress (W)***	0.126	0.084	28266.256	1.511	0.131	-0.038	0.290
Stress (B)	1.157	0.089	22401.612	13.048	<0.001	0.983	1.331

Note. Rows with (W) report within-person effects. Rows with (B) report between-person effects. Asterisks after within-person effects indicate if the variance of the random slopes is significantly greater than zero, suggesting individual differences in fixed effects. SE = standard error; Sig. = significance; CI = confidence interval; LL = lower limit; UL = upper limit; BMI = body mass index; M = male; F = female.

* $p < .05$. *** $p < .001$.

predicted between-person increases in systolic and diastolic blood pressure across the various models and within-person increases in systolic and diastolic blood pressure in some models (see Table 5). The between-person effect of acute stress indicates that people who experienced greater stress also had higher blood pressure than people who experienced less stress.

Question 4: Predicting Threat, Challenge, and Stress From Resting Heart Rate During Sleep

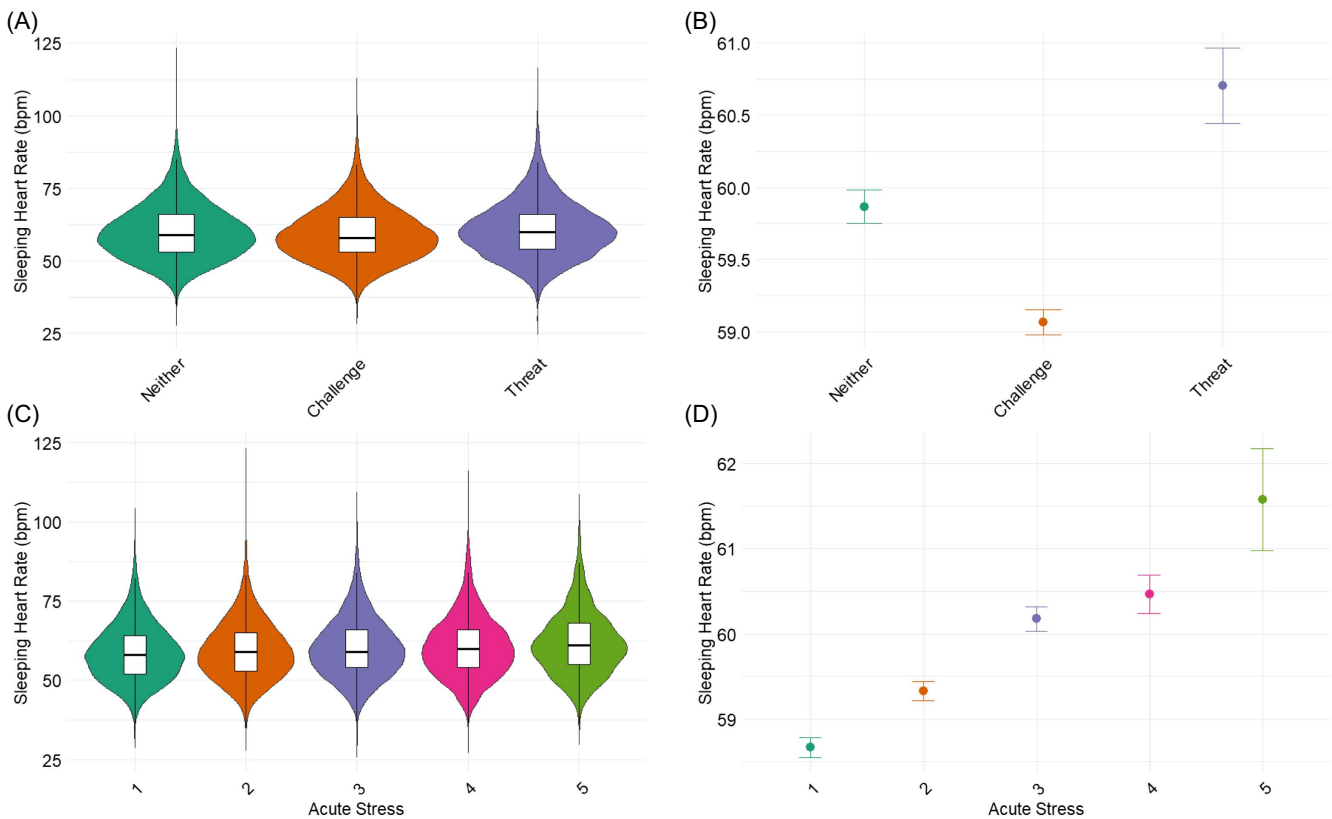
The final step in the analyses was to examine the relationship between resting heart rate during sleep the prior night and subsequent ratings of threat, challenge, and stress. As can be seen in Figure 4, challenge was associated with a decrease in resting heart rate during the prior night's sleep of approximately 3/4 bpm, whereas threat was associated with an increase in resting heart rate during the prior night's sleep of approximately 3/4 bpm. Similarly, each point on the stress scale was associated with approximately a 1/2 bpm increase during the prior night's sleep.

To establish whether resting heart rate during sleep the night before was associated with within- and between-person feelings of threat, challenge, and acute stress, we estimated a series of multilevel models with random slopes and intercepts; threat, challenge, and stress as the dependent variables; resting heart rate as the predictor variable; and controls for age, sex, and BMI. The multilevel models revealed that resting heart rate during sleep the night before was a reliable predictor of within- and between-person changes in threat and challenge, but consistent with the results in Figure 4, the direction of the effect differed for threat and challenge. Increases in resting heart

rate during the prior night's sleep were associated with increased ratings of threat but decreased ratings of challenge. The within-person effects of threat and challenge indicate that on nights prior to an anticipated threat, people's heart rate increased, but on nights prior to an anticipated challenge, people's heart rate decreased, compared to nights where they anticipated no threat or challenge. The between-person effects of threat and challenge indicate that people who anticipated challenge more often had lower heart rates during the night, while people who anticipated threat more often had higher heart rates during the night compared to people who anticipated fewer threats and challenges. Increased resting heart rate during sleep also predicted greater within- and between-person feelings of acute stress (see Table 6). These within-person effects of acute stress indicate that people experienced higher heart rates on nights when they anticipated stressful events compared to nights when they did not. The between-person effects of acute stress indicate that people who anticipated more stressful events had higher heart rates at night than people who anticipated fewer stressful events. All these effects were consistent across the various instantiations of the multilevel models (see online Supplemental Materials).

Discussion

The findings from the current research suggest that when people are awake, threat and challenge are both associated with increased average heart rate and blood pressure. In contrast to these relatively uniform findings while people are awake, threat was associated with increased heart rate during sleep the night before, but challenge was associated with decreased heart rate during sleep the night before.

Figure 4*Violin Plots of Resting Heart Rate During Sleep by Challenge and Threat (Panel A) and Acute Stress Levels (Panel C)*

Note. Each violin plot shows the distribution of heart rate values, with increased width reflecting a greater density of observations. Boxplots within the violins indicate the median and interquartile range of heart rate values for each group. Stress levels were self-reported on a scale of 1–5, where higher values indicate greater stress. Mean estimates with 95% confidence intervals are reported in Panels B and D. See the online article for the color version of this figure.

These data suggest fundamental disconnects between how the body prepares for challenge when people are awake versus asleep and between how people respond to threats and challenges. The body appears to respond adaptively in the face of challenge, reducing heart rate at night prior to anticipated challenges to facilitate restorative sleep but increasing it during the day to address those challenges. In contrast, the body seems incapable of reducing heart rate during sleep when people are faced with upcoming threats, perhaps because of the greater stress associated with threats.

The current findings also raise an unexpected wrinkle with blood pressure, whereby challenge appraisals were associated with a within-person increase in blood pressure but an even larger between-person decrease in blood pressure. We have no evidence pertaining to why this might be, and it is worth keeping in mind that this between-person effect of challenge on blood pressure did not emerge across all the different models. Nevertheless, one possible explanation can be found in the fact that there are stable individual differences in the degree to which people experience threats and challenges (Tomaka & Magoc, 2021). Because people are more likely to take a problem-focused approach to coping with challenges than threats (Hulin et al., 2024), perhaps people who tend to see challenges in their lives are also the sorts who address their problems in a healthier manner. The interpretation and

significance of this unexpected and somewhat unreliable finding clearly remain a question for future research.

Constraints on Generality

There are a number of limitations to this article that might constrain the degree to which it generalizes to other samples or situations. First, our findings may be influenced by the characteristics of the sample, which consisted of individuals who could afford the WHOOP subscription (currently an annual fee of \$239). Their financial status and access to resources might limit the degree to which the results generalize to members of marginalized groups with fewer resources. Thus, an important future goal would be to replicate and extend these results by sampling more broadly from different socioeconomic sectors.

Additionally, although challenge emerged as a more reliable predictor than threat of within-person changes in heart rate and blood pressure, threat and challenge were only measured on three separate occasions and threat was much less commonly experienced than challenge. Had there been more frequent within-person measurements, it is possible that more robust within-person relationships might have emerged between threat, heart rate, and blood pressure. As it stands,

Table 6*Within- and Between-Person Effects of Heart Rate During Sleep on Challenge, Threat, and Acute Stress*

Parameter	Estimate	SE	df	t	Sig.	95% CI	
						LL	UL
Dependent variable: Challenge							
Intercept	0.690	0.016	73049.963	43.987	<0.001	0.659	0.721
BMI	0.000	0.000	72858.430	-0.973	0.331	-0.001	0.000
Age	0.000	0.000	73175.732	-1.229	0.219	0.000	0.000
Sex (M = 0, F = 1)	-0.016	0.004	73123.242	-3.921	<0.001	-0.023	-0.008
HR (W)***	-0.006	0.001	4178.783	-11.932	<0.001	-0.007	-0.005
HR (B)	-0.002	0.000	72614.519	-7.426	<0.001	-0.002	-0.001
Dependent variable: Threat							
Intercept	0.058	0.008	73320.729	7.163	<0.001	0.042	0.074
BMI	0.000	0.000	73173.818	-1.154	0.249	-0.001	0.000
Age	-0.001	8.049E-5	73445.854	-13.033	<0.001	-0.001	-0.001
Sex (M = 0, F = 1)	0.011	0.002	73388.161	5.251	<0.001	0.007	0.015
HR (W)***	0.001	0.000	5389.783	4.287	<0.001	0.001	0.002
HR (B)	0.001	0.000	72946.540	8.368	<0.001	0.001	0.001
Dependent variable: Acute stress							
Intercept	1.982	0.032	71625.618	61.242	<0.001	1.919	2.045
BMI	0.004	0.001	71436.617	4.703	<0.001	0.002	0.005
Age	-0.009	0.000	71682.842	-27.575	<0.001	-0.009	-0.008
Sex (M = 0, F = 1)	0.084	0.008	71690.470	10.191	<0.001	0.068	0.100
HR (W)***	0.007	0.001	3867.383	5.726	<0.001	0.004	0.009
HR (B)	0.007	0.000	71272.765	15.485	<0.001	0.006	0.008

Note. Rows with (W) report within-person effects. Rows with (B) report between-person effects. Stress levels were self-reported on a scale of 1–5, where higher values indicate greater stress. Asterisks after within-person effects indicate if the variance of the random slopes is significantly greater than zero, suggesting individual differences in fixed effects. *SE* = standard error; *Sig.* = significance; *CI* = confidence interval; *LL* = lower limit; *UL* = upper limit; *BMI* = body mass index; *M* = male; *F* = female; *HR* = heart rate.

*** $p < .001$.

the current data suggest that the body prepares for challenge in a more adaptive manner than it does for threat, possibly due to the greater stress associated with threat. In so doing, the results raise the possibility that interventions designed to help people reframe threats as challenges (e.g., Alter et al., 2010) might enable them to achieve greater restorative sleep in anticipation of important future events.

Finally, it should be noted that although the main findings emerged reliably across the various instantiations of the multilevel models, the cardiovascular effects of threat and challenge tended to be small in nature. Perhaps the threats and challenges that people faced during the course of this study were relatively minor, suggesting that it will be important for future research to assess whether the effect sizes remain small when people face threats and challenges that are of great personal importance.

Despite these potential constraints on generality, we believe the effects should be robust to alternative measures of threat and challenge, the inclusion of people of reduced economic means, the use of different biometric capture devices, or any of a variety of other changes in methods and materials. In the absence of evidence to the contrary, we hypothesize that the relationships we document here will replicate regardless of such changes (assuming sufficient power to document relatively small effects).

Additional Limitations

An additional limitation to the current data is that they do not address *why* challenge and threat showed different relationships with heart rate as a function of whether people were asleep or awake.

One clear possibility is that threats are sufficiently stressful to disrupt the parasympathetic dominance that is associated with relaxation and lower resting heart rate during sleep (Trinder et al., 2001). Consistent with this possibility, the within- and between-person coefficients predicting acute stress from threat were two to three times larger than the coefficients predicting acute stress from challenge. These results suggest that the body prepares for difficult anticipated events by reducing heart rate during sleep, but that stress disrupts the body's capacity to reduce heart rate during sleep when anticipated events threaten to overwhelm people.

Another limitation to this study was its exploratory nature and lack of preregistration. This concern is offset to some degree by the large sample size and consistency of the primary results across various analytic models, but future research should address these questions via preregistered research plans. Future research should also manipulate threats and challenges or rely on naturally occurring discontinuities to address the endogeneity problems that make causal inference difficult in the type of research we report here. Because we neither manipulated nor measured the specific challenges and threats that people faced, we do not know whether unmeasured variables might have accompanied them and accounted for their apparent effects. Diary and experience sampling methodologies could address this issue, even if they would not have the power that was enabled by our simpler and less demanding biometric approach.

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